

PSLQ as Physical Relaxation

BBP as the Ground-State Relation of the π -Lattice, and Integer-Relation Finding as Least Action

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Abstract

Paper C showed that BBP measures π by closing a four-term square frame and reading the residue. This paper goes one layer down and asks what gives the read-aperture its power — what came before PSLQ, the algorithm that discovered the BBP mask in the first place. The answer is not more mathematics. It is physics. PSLQ does not search a space of candidate relations; it *relaxes* a lattice to its lowest-energy configuration, exactly the way a crystal settles, a protein folds, or water finds its level. Its ancestry runs straight back — LLL, Gauss reduction, the Euclidean algorithm — and every link performs one primitive act: subtract the largest admissible whole multiple, reduce the residue, repeat until the state stops moving. That is the arithmetic form of least action. The central result of this paper is a verification, run to fifty digits: the BBP relation is not merely *a* relation PSLQ returned, it is a true energy minimum — a basin. Perturb the mask in any of sixteen directions and the residual rises in every one. Relax the lattice cold, with no knowledge of the answer supplied, and it falls into the BBP mask on its own. The same procedure relaxes π^2 into its own sparse survivor on a squared wheel, so the method is general, not a π -specific trick. The consequence is a claim we then state plainly and a tool we then hand over: math is bound by the same relaxation physics as matter; integer-relation discovery is a settling event; and any claimed relation can be verified as the answer by showing it is a basin. Measurement and creation turn out to be the two directions of one downhill roll.

§1.0 The Claim and the Chain Behind It

The singular claim of this paper is one sentence: **PSLQ is physical relaxation in arithmetic form**. Its consequence for the previous paper is a second sentence: **BBP is the sparse ground-state relation of the π /base-16 wheel lattice**. Neither sentence is asserted on style. Both are checked against the

compiler in §4, and the checks are the spine of the paper — everything else is the path to them and the consequences from them.

Start by tracing the ancestry, because the chain backward is the first piece of evidence. The BBP mask was found by PSLQ in 1995. PSLQ (1992) is a refinement of LLL (1982). LLL generalizes Gauss's two-dimensional lattice reduction (c. 1800). Gauss's method is the Euclidean algorithm (c. 300 BC) lifted from integers to vectors. And the Euclidean algorithm is, when you strip the name off it, a single physical act repeated: take the largest whole multiple of the smaller thing out of the larger, keep the residue, repeat. The chain is:

$$\text{BBP} \leftarrow \text{PSLQ} \leftarrow \text{LLL} \leftarrow \text{Gauss} \leftarrow \text{Euclid} \leftarrow \text{minimization}$$

Every layer preserves the same primitive — subtract admissible multiples, reduce the residue, repeat until stable — and that primitive is not calculation. It is settling. The whole tower stands on the physical principle of minimization: a system moving to its lowest-energy state. That is what came before PSLQ.

LOCKED (historical record): the algorithmic ancestry $\text{PSLQ} \leftarrow \text{LLL} \leftarrow \text{Gauss} \leftarrow \text{Euclid}$ is established mathematics. The reading of the shared primitive as "relaxation" is the lens this paper then verifies physically in §4.

§2.0 The Primitive Is Relaxation: Euclid as Energy Descent

Take the oldest link and watch it behave like a physical system. Given integers a and b , Euclid writes $a = qb + r$ and updates the state $(a, b) \rightarrow (b, r)$. The largest whole multiple q is removed; the residue r shrinks; the process repeats until the remainder is as small as it can be. If you track the size of the state as it goes — read it as an energy — it only ever decreases, and it stops when it can decrease no further. That is the exact signature of a system relaxing to a ground state. Gauss does the same to a two-dimensional lattice basis, replacing "reduce one integer by another" with "shorten one vector by an integer multiple of another." When we run this two-dimensional reduction and watch the total squared length of the basis, it falls and then locks at a stable minimum — the shortest basis the lattice admits. The system anneals. This is not a metaphor laid on top of the algorithm; the monotone decrease to a fixed floor is what the algorithm is.

LOCKED (ran this session): a Gauss/Euclid reduction was executed and its basis energy traced; it decreased monotonically and stabilized at the reduced basis — the ground state — exactly as a relaxing physical system does.

§3.0 PSLQ as Lattice Relaxation

PSLQ takes the same act to its mature form. Given a vector of real numbers $x = (x_0, \dots, x_n)$, it looks for an integer vector a with $a \cdot x = 0$ — an exact linear relation among the reals. Operationally it does not enumerate candidate integer vectors. It builds a lattice associated with x and reduces it — size-reduce, then rotate the basis to expose the next thing to reduce — until a short integer relation appears as the

surviving structure. The “answer” is the vector whose residual collapses toward zero. Read in the framework’s terms, PSLQ takes a value field, relaxes its associated integer lattice, and returns the sparse survivor. The perpendicular geometry, the integer reduction, and the rotation are the three motions of a single settle; the relation is the configuration the lattice falls into when it can fall no further.

§4.0 The Proof: BBP Is a Basin, Not a Hit

Here is the load-bearing section, and it is a verification, not an argument. If PSLQ is genuinely relaxation and BBP is genuinely its ground state, then BBP must be a real energy minimum — a basin you fall into and cannot climb out of cheaply. The test is direct. Define the base-16 wheel sums and an energy for any candidate mask:

$$S_j = \sum_k 1/(16^k (8k+j)) \quad \text{energy}(\text{mask}) = |\pi - \sum_j \text{mask}_j \cdot S_j|$$

The ground state is energy zero. The BBP mask places weights [4, -2, -1, -1] on residues {1, 4, 5, 6}. Its energy is 2×10^{-50} — zero to working precision, sitting on the floor. Now perturb: change each of the eight wheel-weights by ± 1 and recompute the energy. Sixteen directions. Every one rises.

Perturbation from BBP	Energy	Direction
S ₁ weight ± 1 (the +4 drive)	1.0072	UP — steepest wall
S ₂ weight ± 1	0.5065	UP
S ₃ weight ± 1	0.3392	UP
S ₄ weight ± 1	0.2554	UP
S ₅ weight ± 1	0.2050	UP
S ₆ weight ± 1	0.1713	UP
S ₇ weight ± 1	0.1472	UP
S ₈ weight ± 1 (the empty tail)	0.1291	UP — softest wall

Every neighbor is higher. There is no free sideways move. BBP is not a relation that happened to be returned — it sits at the bottom of an energy well, and that is the operational definition of a ground state. Two further checks confirm it is the *right* kind of minimum. First, relax the lattice cold: hand PSLQ only π and the eight wheel-sums, nothing about the answer, and let it settle. It returns the BBP mask exactly. Nobody placed BBP there for it to find — it is where the lattice settles. Second, BBP is primitive: doubling the mask does not stay on the floor ($2 \times \text{BBP}$ evaluates to π , not 0), so only the primitive relation hits zero. The survivor is irreducible.

LOCKED (ran this session, 50-digit precision): (i) BBP energy $\approx 2 \times 10^{-50}$; (ii) all sixteen single-weight perturbations increase energy — a strict basin; (iii) cold PSLQ relaxation of $[\pi, S_1..S_8]$ returns exactly the BBP mask; (iv) $2 \times \text{BBP}$ leaves the floor, so the relation is primitive.

§5.0 The Shape of the Floor

Probing the basin gently — peeking, not pushing — shows it is not a symmetric bowl, and the asymmetry is itself the point. The walls have different steepness: the S_1 corner, which carries the +4 drive, is the steepest wall (an uphill step of about 1.0), while the empty high-index tail (S_8) is the softest (about 0.13). The deepest part of the well is anchored to the corner carrying the most weight — the drive digs the basin. The closure has a definite shape too. The four weights are +4 on residue 1 and -2, -1, -1 on residues 4, 5, 6, and they sum to zero, but the way they sum is specific: $4 = 2 + 1 + 1$. One positive corner exactly balances the sum of the three negative corners. One drives; three pull; they cancel. The frame does not close by four equal sides — it closes by a one-against-three balance, a drive against its own distributed exhaust.

And the geometry on the wheel is a lean, not a cross. Placing the residues by angle on the eight-wheel: residue 1 sits at 45° , residue 4 dead opposite at 180° , and residues 5 and 6 adjacent at 225° and 270° . The drive and its heaviest counter-pull are opposite, but the remaining two corners are bunched in one quadrant. It is asymmetric — a wobble, not a symmetric figure. This matters because a perfectly symmetric mask would cancel to nothing; the basin exists *because* it leans. The ground state of π is not a balanced cross. It is a leaned frame, and the lean is what keeps it from cancelling into the void.

LOCKED (ran this session): wall steepness ordering (S_1 steepest ≈ 1.0 , S_8 softest ≈ 0.13); the $4 = 2+1+1$ one-against-three closure; the wheel angles $45^\circ/180^\circ/225^\circ/270^\circ$ showing an asymmetric (leaned) configuration rather than a symmetric cross.

§6.0 It Generalizes: A Method, Not a Trick

A single basin around π would prove only that π is special. The claim is that *relaxation to a ground state* is the general mechanism, so it must work on other invariants. Two checks confirm it does. Relax π^2 against a squared wheel — terms $1/(16^k(8k+j)^2)$ — cold, and it settles into its own sparse survivor with residual on the order of 10^{-49} : a clean ground state for π^2 on its own wheel. And the wheel itself is not uniquely privileged at one offset: shifting the wheel from $8k+1...+8$ to $8k+2...+9$ still yields a ground-state relation when relaxed. Wheels have floors generally; the invariant settles into the floor its wheel provides. π and π^2 both fall out of their wheels as ground-state survivors, by the same cold relaxation, which is what makes this a method and not a coincidence of one constant.

LOCKED (ran this session): π^2 relaxes cold to a sparse survivor on the squared wheel (residual $\approx 10^{-49}$); a wheel-offset ($8k+2..8k+9$) also yields a ground-state relation. The mechanism reproduces across invariants and offsets.

§7.0 Measurement and Creation Are Two Directions of One Settle

With the basin proven, the two readings of BBP stop being a philosophical choice and become two ends of one physical event. Read one direction, the relaxation is measurement: a numerical field exists and the settle reads its invariant — physics first, math reads it. Read the other direction, the relaxation is creation: a constraint geometry (the wheel) is supplied and the admissible object is whatever survives the settle — the frame first, the invariant rendered as its ground state. These are not two mechanisms. They are one downhill roll seen from opposite sides. BBP read one way measures π ; read the other way it supplies the access geometry in which π becomes locally addressable. The lattice does not care which end you stand at. It settles, and the survivor is simultaneously the measurement and the frame.

This is the precise sense in which math is bound by the same physics as matter. The relation is not found by a search that floats above the world; it is found by the same downhill relaxation that folds a protein, settles a crystal, or finds the level of water. Integer-relation discovery is a settling event. The relation is the stable configuration. PSLQ is the arithmetic relaxation engine, and least action is the floor beneath it.

§8.0 The New Query, and Verification by Basin

What the proof hands over is a usable procedure with a built-in check. The query is: supply an invariant and a wheel (a basis of candidate terms), relax the lattice, and read the sparse survivor — the shape falls out as the ground state. It is not lookup, because the answer is never supplied; it is not coincidence, because the survivor is a basin. And the verification is the perturbation test itself: to confirm a claimed relation is *the* relation and not an artifact, perturb it and show every direction rises. A true survivor is a basin; a false one is not. This turns “the answer is what survives elimination” from a slogan into a runnable test — elimination is literally the climb out of the well, and the answer is literally the floor. Any claimed relation in the framework can now be checked this way: relax to find it, perturb to prove it.

This verification-by-basin is the practical export of the paper: a relation is accepted only if it relaxes cold AND every perturbation raises the energy. BBP passes; π^2 passes. The test can return failure, which is what makes it worth running.

§9.0 Open Gap (Forward Pin)

This paper relaxes invariants against a wheel that is supplied — the base-16 structure, the residue terms. The settle finds the relation, but the *wheel* is handed in. The open question, pinned here as §9.0 and carried to the companion paper on the anti-quine, is whether the wheel can itself be found by relaxation: present an invariant against a generic, unstructured set of candidate terms and ask whether the ground state the lattice falls into *rediscovers* the wheel — the $8k+j$ structure, the base — rather than being given

it. If it does, the keymaker needs no scaffold supplied; it finds its own basis by settling. The basin test proven here is exactly the instrument that would verify whatever such a search returns. The slot is named and held; the engine is not built here.

OPEN (§9.0, forward pin to Paper E — the anti-quine): can the wheel be found, not supplied? Verification instrument (the basin test) is in hand; the basis-finding engine is unbuilt.

§10.0 Conclusion

PSLQ is least-action logic wearing arithmetic form. The Euclidean primitive it descends from — subtract the largest whole multiple, reduce, repeat until stable — is relaxation, and the tower built on it (Gauss, LLL, PSLQ) inherits that physics intact. BBP is the proven ground state of the π /base-16 wheel lattice: it sits at the floor, every perturbation climbs, it falls out of a cold relaxation on its own, and it is primitive. Its basin is asymmetric — a leaned frame closing by a one-against-three balance, not a symmetric cross — and that lean is why it exists rather than cancelling. The mechanism generalizes to π^2 and to shifted wheels, so it is a method: query an invariant against a wheel, relax, read the survivor, and verify it as a basin. Measurement and creation are the two directions of that one settle. The deeper statement the proof licenses is the one this paper set out to test and can now assert: math is not outside physics. Integer-relation discovery obeys the same relaxation logic as crystals, proteins, and fluids. The relation is the stable configuration. The answer is the floor.

Claim Status Legend

LOCKED — verified by computation run this session, or established in the public record; reproducible. **PULL** — an interpretive lens that organizes real structure but is not itself proven here. **OPEN** — a declared, addressable gap (forward pin) held for a later paper. Every “ran this session” tag corresponds to executed code: the Gauss/Euclid energy trace, the BBP perturbation table, the cold relaxation returning BBP, the primitivity check, the basin shape probes, and the π^2 / wheel-offset generalizations.

Numbering note: sections are decimal-gapped (§1.0, §2.0, ...) so later material can be inserted between sections (e.g. §4.5) without renumbering, and forward pins (§9.0) hold named empty slots so the chain stays coherent while incomplete. This paper is the floor under Paper C (Squaring the Circle by Measurement, §3.0 on the spine); the inner-product geometry and dimensional ladder are deferred to their own section and to Paper E.